

$$m v^2 r^{-1} = F_c = q \vec{v} \times \vec{B}$$

$$m \omega^2 r = \vec{p} \cdot \omega_c$$

$$\omega^2 r = v^2 r^{-1}$$

$$\omega^2 = \frac{v^2}{r^2}$$

$$\omega = \frac{v}{r}$$

$$\frac{q B r}{m v} \Rightarrow \omega = \frac{q B}{m}$$

For relativistic particles:

$$\gamma \vec{p} \omega_c = F_c \Rightarrow \frac{q B}{\gamma m} = \omega_c \xrightarrow{\text{cyclotron freq.}} \xrightarrow{\text{period}} T_c = \frac{\gamma m}{q B}$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow \text{Lorentz factor}$$

$$R_c = \frac{\gamma m \vec{v}}{q B} = \frac{p_{\perp}}{q B} \rightarrow \text{perpendicular momentum}$$

(radius of curvature)

AMS uses: Measure curvature $\rightarrow$ get $p_{\perp}$

$\rightarrow$ get rigidity = $p/q \rightarrow p = \frac{p_{\perp}}{\sin \theta}$, θ is the angle between $\vec{B}$ and $\vec{v}$.

$$E^2 = p^2 + m^2 \text{ (energy in GeV)}$$

$$\gamma = \frac{E}{m}, \beta = \frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}}$$

lorentz-rhs $\rightarrow$ System of ODEs: (simple-track.ipynb)

$$\frac{d\vec{r}}{dt} = \vec{v}, \quad \frac{d\vec{v}}{dt} = \frac{q}{m} \vec{v} \times \vec{B}$$

solve_ivp from SciPy solves this system.

(in the function simulate-trajectory)

fit_circle_least_squares $\rightarrow$ finds circular fit
for x and y : $x^2 + y^2 + Ax + By + C = 0$

where: $x_c(a\text{-param}) = -A/2$

$y_c(b\text{-param}) = -B/2$

$C\text{-param} = -C$

$$R = \sqrt{x_c^2 + y_c^2 - C} = R_c = \frac{p_{\perp}}{q \vec{B}}$$

estimate-momentum-from-trajectory
finds rigidity = $p_{\perp}/q$
 $= R_c \vec{B}$

* Simulating proton/antiproton events.

rigidity-to-relativistic: $\rightarrow$ takes in charge, mass
and rigidity.

$\rightarrow$ returns beta, gamma
and momentum.

velocity-from-speed-and-pitch: Dissects speed
to velocity magnitudes
in parallel and perp.
directions.

Boris pusher: Rotates $\vec{v}$ around $\vec{B}$ by small
angle steps, representing: $\frac{d\vec{v}}{dt} = \frac{q}{\gamma m} \vec{v} \times \vec{B}$

This replaces the system of Lorentz ODE.

simulate-relativistic-track $\rightarrow$ the main function.

$\rightarrow$ can simulate any hadronic/leptonic particle with Z , mass-GeV and mass-kg.

$\rightarrow$ rigidity and pitch angle are chosen as well (later randomized in silicon-tracker-simulation)

$\rightarrow$ derives γ , β and $p_{\perp}$ from rigidity/angle.

$\rightarrow$ Computes cyclotron freq./period ($\omega_c(T_c)$)

$\rightarrow dt = T_c / \text{steps}$ (200 is chosen.)
(per period)

$\rightarrow$ Applies Boris pusher, returns t , $\vec{r}$, $\vec{v}$, γ .

Some circle fit with simple-track.

* silicon-tracker-simulation

$\rightarrow$ uses relativistic-model-track as physics engine.

$\rightarrow$ 7 planes within 0.1m and 0.9m, similar to AMS L2-L8 inner layers.

intersect-track-with-layers: Uses trilinear interpolation between points from simulated track to find points of intersection.

$$\rightarrow X_{\text{hit}} = X_1 + \alpha(X_2 - X_1) \quad \alpha = \frac{Z_1 - Z_0}{Z_2 - Z_1}$$

$$Y_{\text{hit}} = Y_1 + \alpha(Y_2 - Y_1)$$

$$y_{hit} = y_1 + \alpha (y_2 - y_1)$$

$$z_{hit} = z_L \text{ (already known)}$$

~ random noise is added with $\sim N(0, 50\text{mm})$.

~ Then data is written on pandas dataframe.